

# RoboRetail

Web app URL: [RoboRetail Transaction Management System](#)

## **Executive Summary**

During the review stages of our designs we received valuable feedback from peers and instructional staff and implemented several of the improvement suggestions throughout the term. The changes include adding a cost attribute to the ProductSuppliers table, adding subqueries to the insert sample data, revamping of our ERD, and fixing user interface functionality.

Our most impactful design change revolves around the handling of foreign keys. We had initially utilized cascade with an isActive flag for our Customers and Suppliers tables to allow data retention and prevent loss of historical data. However, during the final step of our review and last implementation stage, we decided against the usage of the restrict to focus solely on the transactional aspects of the design. Doing so keeps queries fast and responsive, while maintaining foreign key values and ensuring accuracy in other entities. Without disregarding the potential loss of data with a hard delete, our database universe assumes a historical data preservation outside of the operational database.

As a result, the current implementation shows a prioritization of performance and simplifying maintenance, and staying consistent with database transaction systems. Our team believes this implementation choice aligns best with what the course project intends to teach.

Our team utilized AI for several different purposes, including brainstorming, generating SQL queries and stored procedures, and assisting with setting up the webpage connection

to our database. The usage of Ai increased during the later stages of development as the workload became more time consuming and repetitive.

The only weakness with AI we noticed is with continuous usage or re-prompting the AI can get lost or confused and make up data. When working with AI our team focused on a small aspect of the project and helped abstract away as much of the non-required information. Doing so resulted in an AI assistant that was focused on a hyperspecific goal and target, without getting overloaded with irrelevant information.

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## **Overview**

RoboRetail is an online general retail store founded in 2020 during COVID lockdown that processes approximately 2,500 online customer orders per month, generating roughly \$1 million in annual sales. As order volume has increased, RoboRetail has experienced operational challenges related to manually tracking customer transactions, product inventory levels, and supplier relationships across its growing catalog of retail goods.

Without a centralized database system, the company faces issues such as inaccurate inventory counts, delayed order fulfillment, and difficulty identifying which suppliers provide specific products when restocking inventory. These inefficiencies can result in order processing delays, overselling out-of-stock items, and increased administrative effort when managing customer and supplier information.

To address these problems, RoboRetail requires a database-driven website that records customer Orders, tracks the Products included in each Order, monitors inventory availability, and maintains Supplier information for product resupply. The proposed Transaction Management System will support accurate order processing by linking Customers to their Orders, associating Orders with the Products purchased, and

maintaining Supplier relationships for each Product to streamline inventory replenishment and fulfillment operations.

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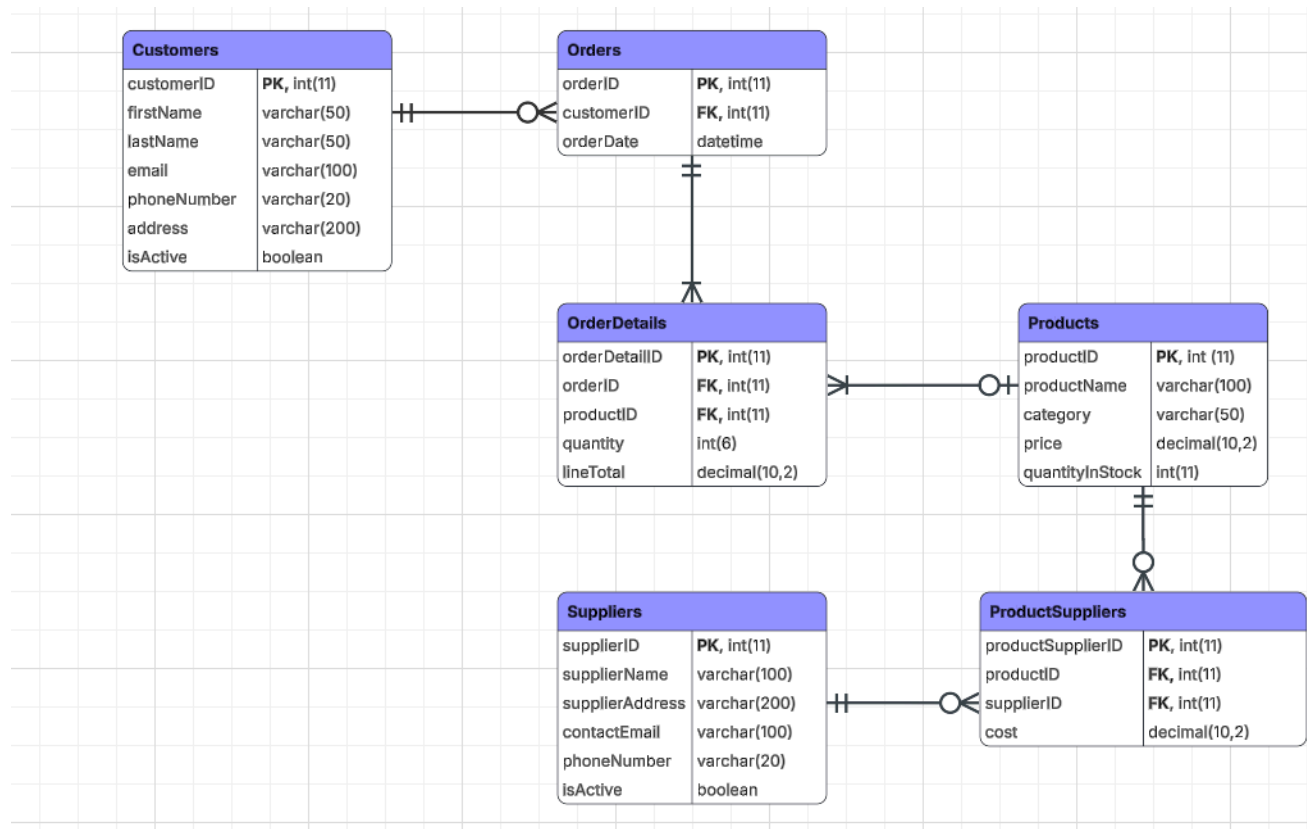
## **Database Outline**

- **Customers:** Stores customer information used to track orders and manage relationships.
  - customerID: int, auto\_increment, unique, not NULL, PK
  - firstName: varchar(50), not NULL
  - lastName: varchar(50), not NULL
  - email: varchar(100), unique, not NULL
  - phoneNumber: varchar(20), NULL
  - address: varchar(200). NULL
  - isActive: boolean, default True
    - **Relationship:**
      - Each customer can place multiple orders,
      - Each order belongs to only one customer.
      - A 1:M relationship between Customers and Orders is implemented with customerID as a FK inside of Orders.
  
- **Products:** Stores information about items available for sale and inventory tracking.
  - productID: int, auto\_increment, unique, not NULL, PK
  - productName: varchar(100), not NULL
  - category: varchar(50), not NULL
  - price: decimal(10, 2), not NULL
  - quantityInStock: int, not NULL
    - **Relationship:**
      - Products appear in multiple orders through a many-to-many (M:N) relationship with Orders using OrderDetails.
      - Products can be supplied by multiple suppliers using many-to-many (M:N) relationship via ProductSuppliers.
  
- **Orders:** Records each sales order placed by customers
  - orderID: int, auto\_increment, unique, not NULL, PK
  - customerID: int, not NULL, FK references Customers(customerID)
  - orderDate: datetime, not NULL

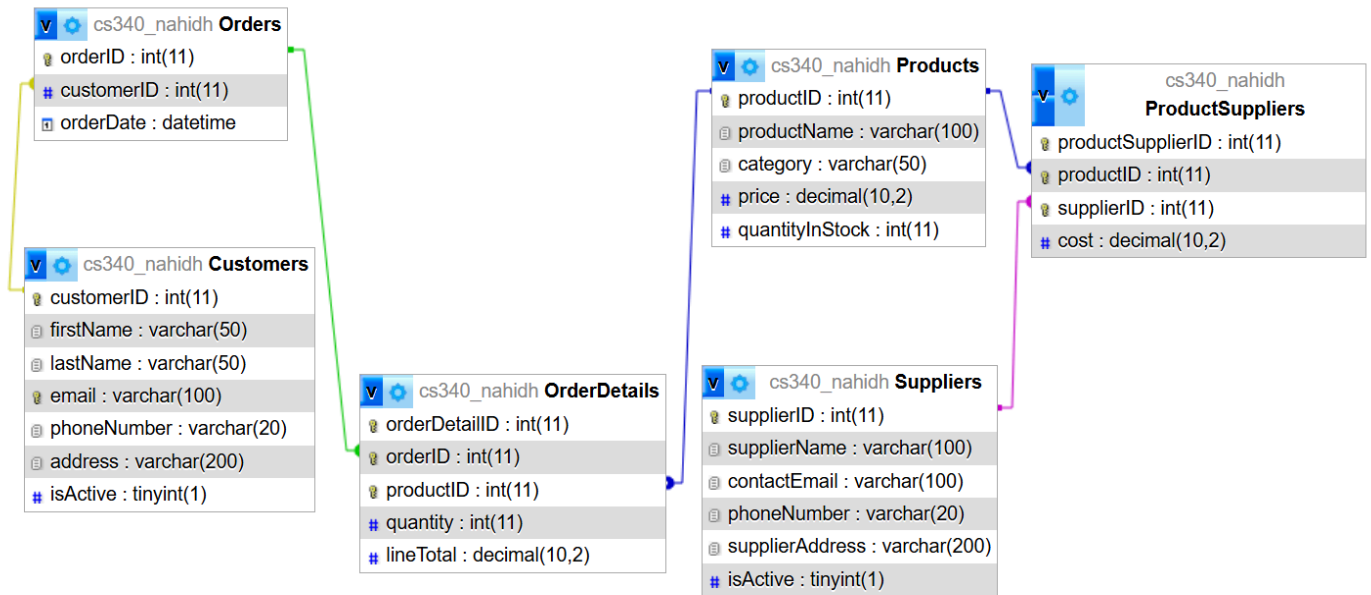
- **Relationship:**
    - Each order belongs to one customer(1:M with Customers).
    - Orders can include multiple products(M:N with Products via OrderDetails), as each order may include multiple products and each product may appear in multiple orders.
- **OrderDetails:** Captures the products included in each order, including quantity and total price per item.
  - orderDetailID: int, auto\_increment, unique, not NULL, PK
  - orderID: int, not NULL, FK references Orders(orderID)
  - productID: int, not NULL, FK references Products(productID)
  - quantity: int, not NULL
  - lineTotal: decimal(10, 2), not NULL
    - **Relationship:**
      - Each record links one order to one product.
      - Allows tracking multiple products per order.
- **Suppliers:** Stores information about suppliers providing products.
  - supplierID: int, auto\_increment, unique, not NULL, PK
  - supplierName: varchar(100), not NULL
  - supplierAddress: varchar(200), not NULL
  - contactEmail: varchar(100), NULL
  - phoneNumber: varchar(20), NULL
  - isActive: boolean, default True
    - **Relationship:**
      - Suppliers can supply multiple products( M:N with Products via ProductSuppliers)
- **ProductSuppliers:** Resolves the many-to-many relationship between Products and Suppliers and tracks supplier-specific product costs.
  - productSupplierID: int, auto\_increment, unique, not NULL, PK
  - productID: int, not NULL, FK references Products(productID)
  - supplierID: int, not NULL, FK references Suppliers(supplierID)
  - cost: decimal(10, 2), not NULL.
    - **Relationship:**
      - Each record links one product to one supplier.
      - Allows multiple suppliers for each product and multiple products for each supplier.

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## Entity-Relationship Diagram:



## Schema Diagram:



## Sample Data

### Customers

*Stores customer information.*

| customerID | firstName | lastName | email              | phoneNumber  | address               | isActive |
|------------|-----------|----------|--------------------|--------------|-----------------------|----------|
| 1          | John      | Doe      | john@email.com     | 123-456-7890 | 123 Main St           | 1        |
| 2          | Jane      | Smith    | jane@email.com     | 234-567-8901 | 456 Oak St            | 1        |
| 3          | Alex      | Lee      | alex@email.com     | NULL         | 789 Pine St           | 1        |
| 4          | Cardinal  | Ericsson | cardinal@email.com | 708-739-9294 | 2387 Agriculture Lane | 1        |
| 5          | Curtis    | Taylor   | curtis@email.com   | 317-223-6447 | 3026 Henery Street    | 1        |

### Products

*Stores product and inventory information.*

| productID | productName | category    | price  | quantityInStock |
|-----------|-------------|-------------|--------|-----------------|
| 1         | Laptop      | Electronics | 999.99 | 10              |
| 2         | Headphones  | Electronics | 199.99 | 25              |
| 3         | Desk Chair  | Furniture   | 150.00 | 8               |
| 4         | Smartphone  | Electronics | 899.99 | 15              |
| 5         | Pillow      | Bedding     | 19.66  | 5               |

## Orders

Stores customer orders.

| orderID | customerID | orderDate           |
|---------|------------|---------------------|
| 1       | 1          | 2026-04-01 10:00:00 |
| 2       | 2          | 2026-04-02 12:30:00 |
| 3       | 1          | 2026-04-03 09:15:00 |
| 4       | 3          | 2026-04-04 08:30:00 |
| 5       | 5          | 2026-04-12 18:35:00 |

Demonstrates:

- Customer 1 has multiple orders (1:M relationship)

## OrderDetails

Links Orders and Products (M:N relationship).

| orderDetailID | orderID | productID | quantity | lineTotal |
|---------------|---------|-----------|----------|-----------|
| 1             | 1       | 1         | 1        | 999.99    |
| 2             | 1       | 2         | 2        | 399.98    |
| 3             | 2       | 3         | 1        | 150.00    |
| 4             | 3       | 5         | 2        | 39.32     |
| 5             | 4       | 4         | 1        | 899.99    |

Demonstrates:

- One order contains multiple products
- One product appears in multiple orders



## Suppliers

Stores supplier information.

| supplierID | supplierName | contactEmail           | phoneNumber  | supplierAddress |
|------------|--------------|------------------------|--------------|-----------------|
| 1          | TechCorp     | tech@corp.com          | 111-222-3333 | 12 Tech Ave     |
| 2          | FurniCo      | contact@furni.com      | 444-555-6666 | 45 Wood St      |
| 3          | Sleep City   | support@sleep-city.com | 127-893-5582 | 750 Gerald Ave  |

## ProductSuppliers

Links Products and Suppliers (M:N relationship) and tracks cost.

| productSupplierID | productID | supplierID | cost   |
|-------------------|-----------|------------|--------|
| 1                 | 1         | 1          | 700.00 |
| 2                 | 2         | 1          | 120.00 |
| 3                 | 3         | 2          | 90.00  |
| 4                 | 4         | 1          | 690.00 |
| 5                 | 5         | 3          | 10.33  |

Demonstrates:

- One product has **multiple suppliers**
  - One supplier provides **multiple products**
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UI Pages: Database Reset In navigation section on each page.

## Customers

Customers Table Functionality: Read / Add / Update / Delete

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### Browse Customers

| Customer ID | First Name | Last Name | Email              | Phone #      | Address               |
|-------------|------------|-----------|--------------------|--------------|-----------------------|
| 1           | John       | Doe       | john@email.com     | 123-456-7890 | 123 Main St           |
| 2           | Jane       | Smith     | jane@email.com     | 234-567-8901 | 456 Oak St            |
| 3           | Alex       | Lee       | alex@email.com     |              | 789 Pine St           |
| 4           | Cardinal   | Ericsson  | cardinal@email.com | 708-739-9294 | 2387 Agriculture Lane |
| 5           | Curtis     | Taylor    | curtis@email.com   | 317-223-6447 | 3026 Henery Street    |

## Add / Update / Delete

### Add Customer

First Name:

Last Name:

Email:

Phone Number:

Address:

[Add Customer](#)

### Update Customer

Select Customer:

Updated Email:

Updated Phone #:

Updated Address:

[Update Customer](#)

### Delete Customer

Select Customer:

[Delete Customer](#)

# Products

Products Functionality: Read / Add / Update / Delete

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### Browse Products

| Product ID | Product Name | Category    | Price  | Quantity in Stock |
|------------|--------------|-------------|--------|-------------------|
| 1          | Laptop       | Electronics | 999.99 | 10                |
| 2          | Headphones   | Electronics | 199.99 | 25                |
| 3          | Desk Chair   | Furniture   | 150.00 | 8                 |
| 4          | Smartphone   | Electronics | 899.99 | 15                |
| 5          | Pillow       | Bedding     | 19.66  | 5                 |

## Add / Update / Delete

### Add Product

Product Name:

Category:

Price:

Quantity In Stock:

[Add Product](#)

### Update Product

Select Product:

Updated Name:

Updated Category:

Updated Price:

Updated Qty:

[Update Product](#)

### Delete Product

Select product:

[Delete Product](#)

# Orders

Orders Functionality: Read / Add / Update / Delete

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Browse Orders

| Order ID | Customer ID | Customer First Name | Customer Last Name | Order Date                                                |
|----------|-------------|---------------------|--------------------|-----------------------------------------------------------|
| 1        | 1           | John                | Doe                | Wed Apr 01 2026 10:00:00 GMT-0700 (Pacific Daylight Time) |
| 2        | 2           | Jane                | Smith              | Thu Apr 02 2026 12:30:00 GMT-0700 (Pacific Daylight Time) |
| 3        | 3           | Alex                | Lee                | Fri Apr 03 2026 09:15:00 GMT-0700 (Pacific Daylight Time) |
| 4        | 4           | Cardinal            | Ericsson           | Sat Apr 04 2026 08:30:00 GMT-0700 (Pacific Daylight Time) |
| 5        | 5           | Curtis              | Taylor             | Sun Apr 12 2026 18:35:00 GMT-0700 (Pacific Daylight Time) |

## Add / Update / Delete

Add Order

Customer: 1 - John

Order Date: mm/dd/yyyy --:-- --

Add Order

Update Order

Select Order: 1 - John Doe

Updated Order date: mm/dd/yyyy --:-- --

Update Order

Delete Order

Select Order: 1 - John Doe

Delete Order

# Order Details

Order Details Functionality: Read

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Browse Order Details

| Order Detail ID | Order ID | Product ID | Product Name | Quantity | Line Total |
|-----------------|----------|------------|--------------|----------|------------|
| 1               | 1        | 1          | Laptop       | 1        | 999.99     |
| 2               | 1        | 2          | Headphones   | 2        | 399.98     |
| 3               | 2        | 3          | Desk Chair   | 1        | 150.00     |
| 4               | 3        | 5          | Pillow       | 2        | 39.32      |
| 5               | 4        | 4          | Smartphone   | 1        | 899.99     |

# Suppliers

Suppliers Functionality: Read / Add / Update / Delete

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Browse Suppliers

| Supplier ID | Supplier Name | Contact Email          | Phone #      | Address        |
|-------------|---------------|------------------------|--------------|----------------|
| 1           | TechCorp      | tech@corp.com          | 111-222-3333 | 12 Tech Ave    |
| 2           | FurniCo       | contact@furni.com      | 444-555-6666 | 45 Wood St     |
| 3           | Sleep City    | support@sleep-city.com | 127-893-5582 | 750 Gerald Ave |

## Add / Update / Delete

### Add Supplier

Supplier Name:

Contact Email:

Phone Number:

Supplier Address:

Add Supplier

### Update Supplier

Select Supplier:

Updated Supplier Name:

Updated Contact Email:

Updated Phone Number:

Updated Supplier Address:

Update Supplier

### Delete Supplier

Select Supplier:

Delete Supplier

## Product Suppliers

Product Suppliers Functionality (M:N): Read / Add / Update / Delete

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Reset database

### Browse Product Suppliers

| Product Supplier ID | Product ID | Product Name | Supplier ID | Supplier Name | Cost (\$) |
|---------------------|------------|--------------|-------------|---------------|-----------|
| 1                   | 1          | Laptop       | 1           | TechCorp      | 700.00    |
| 2                   | 2          | Headphones   | 1           | TechCorp      | 120.00    |
| 3                   | 3          | Desk Chair   | 2           | FurniCo       | 90.00     |
| 4                   | 4          | Smartphone   | 1           | TechCorp      | 690.00    |
| 5                   | 5          | Pillow       | 3           | Sleep City    | 10.33     |

## Add / Update / Delete

### Add Product Supplier

**Restriction:** Do not add a duplicate Product ID-Supplier ID combo. Check table above to confirm

**Product:**

**Supplier:**

**Cost:**

Add Product Supplier

### Update Product Supplier

**Restriction:** do not select a pre-existing (Product ID, SupplierID) combo (see table above) unless it pertains to the same Product-Supplier ID. E.g. you can select Product ID = 3 and Supplier ID = 2 only if you also select Product-Supplier ID of 3

**Select Product-Supplier ID (check table above):**

**Select Product ID:**

**Select Supplier ID:**

**Updated Cost:**

Update Product Supplier

### Delete Product Supplier

**Select Relationship:**

Delete Product Supplier

## **Citations**

Honcharuk, Y., & DashDevs | FinTech Software Development Company. (2024, September 25). *Step-by-step guide to project roadmap*. DashDevs.  
<https://dashdevs.com/blog/effective-roadmap-step-by-step/>

*All specific AI prompts are cited in the code files they were utilized.*

OpenAI. ChatGPT (2026), <https://chat.openai.com/>

Anthropic Claude (2026), <https://claude.ai/>

Microsoft. Microsoft Copilot. (2026) <https://copilot.microsoft.com/>